Damon H. T. Cheung

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EDUCATION

University of Michigan, Ann Arbor
Ph.D. in Physics — 2022 – Present

The Chinese University of Hong Kong
B.Sc. in Physics — *Enrichment Stream in Theoretical Physics* — 2018 – 2022

AWARDS & LEADERSHIP

- LIGO-Virgo-KAGRA Continuous Waves Working Group, Directed Searches Subgroup Chair — 2026 – Present
- Frank Sevcik Award — 2026
- Michigan Institute for Computational Discovery & Engineering Fellowship — 2025

RESEARCH EXPERIENCE

Research Assistant, University of Michigan, Ann Arbor – U.S. — Sep 2022 – Present
Advisor: Prof. Keith Riles

- Analyzed large-scale time-series data using semicoherent matched filtering to search for continuous gravitational-wave (CW) signals from supernova remnants and globular clusters, setting the most stringent astrophysical constraints to date on neutron star ellipticity and r-mode instability amplitudes.
- Developing a TwoSpect-based semicoherent matched-filtering search pipeline in JAX for CW signals from neutron stars in binary systems.
- Developed PAWS, a scalable, object-oriented data analysis framework in Python, enabling efficient processing of massive CW datasets.
- Built an Attention U-Net for CW signal detection using PyTorch, achieving sensitivity comparable to state-of-the-art deep-learning searches with substantially less training time and data.

AI/ML Research Intern, Polymathic Technologies Inc., New York – U.S. — May 2026 – Aug 2026

- Conducted research on multimodal foundation models in JAX, extending their capabilities to new data modalities and improving performance on downstream tasks.
- Studied the performance of time-series tokenization schemes on real-world data and integrated them into a foundation model for time-series prediction.

Research Assistant, The Chinese University of Hong Kong – Hong Kong — Dec 2021 – Aug 2022
Advisor: Prof. Otto A. Hannuksela

- Derived the correct form of the strong-lensing Bayes factor for gravitational-wave (GW) event pairs by reweighting parameter-estimation posteriors with the astrophysical population prior.
- Showed that nonparametric population reconstructions yield a smaller Bayes factor bias than parametric population models.

Guest Researcher, Center for Computational Astrophysics, Flatiron Institute – U.S. — May 2021 – Dec 2021
Advisors: Dr. Kaze W. K. Wong and Prof. Shirley Ho

- Performed hierarchical Bayesian inference with machine-learning likelihood emulators for parameter estimation of GW populations from mock and real observation data.
- Benchmarked Gaussian process and normalizing flow emulators for simulation-based inference using high-dimensional GW population models.
- Showed that normalizing flow emulators recover the population posterior on mock catalogs, while Gaussian process emulators are effective only in low-dimensional cases.

Summer Research Assistant, The Chinese University of Hong Kong – Hong Kong — May 2020 – Dec 2020
Advisor: Prof. Ming Chung Chu

- Contributed to the development of SemiCore, a dark matter halo simulation code in Julia, incorporating decaying dark matter (DDM) properties and a generalized gravitational potential.
- Analyzed the impact of decay half-life and recoil velocity on the time evolution of DDM halo density profiles.

TEACHING EXPERIENCE

Teaching Assistant, University of Michigan, Ann Arbor – U.S. Sep 2022 – May 2024

- Physics 405: Electrodynamics
- Physics 453: Quantum Mechanics
- Physics 251: Experimental Physics for the Life Sciences

SELECTED PRESENTATIONS

APS Global Physics Summit, Denver, U.S. 2026

Robustness of Template-Based Search for Continuous Gravitational Waves with Glitches

Hong Kong Laureate Forum, Hong Kong 2025

Attention U-Net for All-Sky Continuous Gravitational Wave Searches

LVK Collaboration Meeting, Fort Collins, U.S. 2025

Attention U-Net for All-Sky Continuous Gravitational Wave Searches

APS Global Physics Summit, Anaheim, U.S. 2025

A Directed Search for Continuous Gravitational Waves from Supernova Remnants in the LIGO O4a Data Using a Semi-Coherent Templated Search Method

Kavli Institute for Astronomy and Astrophysics, Peking University, Beijing, China 2024

Semi-Coherent Search Method for Continuous Gravitational Waves Using WEAVE

LVK Collaboration Meeting, Toyama, Japan 2023

Mitigating the Effect of the Population Model Uncertainty on the Strong Lensing Bayes Factor Using Nonparametric Methods

SELECTED PUBLICATIONS

- **The G347.3-0.5 outlier from O3: a follow-up case study for continuous gravitational-wave candidates**, arXiv:2608.07662 (2026)
  Lorenzo Mirasola, Francesco Amicucci, Marta Carrió Thomás, **Damon H. T. Cheung**, *et al.*
- **Searches for Continuous Gravitational Waves from Supernova Remnants in the first part of the LIGO-Virgo-KAGRA Fourth Observing run**, ApJ 1004, 204 (2026)
  LIGO Scientific, Virgo, and KAGRA Collaborations (**Damon H. T. Cheung** as one of the main analysts)
- **Search for continuous gravitational waves from neutron stars in five globular clusters in the first part of the fourth LIGO–Virgo–KAGRA observing run**, Phys. Rev. D 114, 062001 (2026)
  Damon H. T. Cheung, Keith Riles, *et al.*
- **Determining the Hubble Constant through Cross-Correlation of Galaxies and Gravitational Waves**, Phys. Rev. D 113, 103532 (2026)
  Jiaming Pan, Dragan Huterer, Camille Avestruz, **Damon H. T. Cheung**, Emery Trott, Neal Dalal, and Donghui Jeong
- **Attention U-Net for all-sky continuous gravitational wave searches**, Phys. Rev. D 112, 103039 (2025)
  Damon H. T. Cheung
- **Mitigating the effect of the population model uncertainty on the strong lensing Bayes factor using nonparametric methods**, Phys. Rev. D 111, 063012 (2025)
  Damon H. T. Cheung, Stefano Rinaldi, Martina Toscani, and Otto A. Hannuksela
- **Testing the robustness of simulation-based gravitational-wave population inference**, Phys. Rev. D 106, 083014 (2022)
  Damon H. T. Cheung, Kaze W. K. Wong, Otto A. Hannuksela, Tjonnie G. F. Li, and Shirley Ho

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